

# Continuous functions – 8

Sets of discontinuity, oscillation and the Baire category theorem

Homework

1 сентября 2026 года

**Theoretical minimum.** Oscillation and the  $F_\sigma$  structure of the discontinuity set;  $G_\delta$  sets; the Baire category theorem and its typical applications; functions of the first Baire class; Thomae's function; Lebesgue's criterion for Riemann integrability; sets of measure zero versus meagre sets.

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1. Prove that a pointwise limit of continuous functions is of the first Baire class, and that such a function has a dense  $G_\delta$  set of continuity points; deduce that the Dirichlet function is not a pointwise limit of continuous functions.
2. Prove that a monotone function has at most countably many discontinuities and that for every countable set  $S$  there is a monotone function whose discontinuity set is exactly  $S$ .
3. Prove that the set of points at which a function  $f: \mathbb{R} \rightarrow \mathbb{R}$  has a local maximum with a strict inequality on a neighbourhood is at most countable.
4. Prove that there is no sequence of continuous functions converging pointwise to the indicator of the rationals, and explain how this follows from Theorem 2 of the class sheet.
5. Prove that a set  $E \subseteq \mathbb{R}$  is both meagre and of full measure for some choice of  $E$ ; that is, meagreness and null measure are genuinely different notions.

6. Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be such that for every  $x$  the limit  $\lim_{y \rightarrow x} f(y)$  exists. Prove that the set of points where this limit differs from  $f(x)$  is at most countable.
7. Prove that a function which is continuous at every point of a dense set need not be Riemann integrable, and give an example of a Riemann integrable function whose discontinuity set is dense.
8. Prove that if  $f_n \rightarrow f$  pointwise on a complete metric space and each  $f_n$  is continuous, then for every  $\varepsilon > 0$  there is a non-empty open set on which the convergence is uniform up to  $\varepsilon$  (Osgood's theorem).
9. Prove that the Cantor set is nowhere dense, of measure zero and of cardinality continuum, and construct a continuous function which is constant on every complementary interval and yet non-constant (the Cantor staircase).
10. Prove that the set of nowhere differentiable functions is residual in  $C[0, 1]$ , and explain why this gives an existence proof for the Weierstrass function without writing it down.